

LESSON 5-3 • SESSION 1 • PRACTICE • FORM A

Trapezoids by Decomposing

Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles. · 6.G.A.1

REMEMBER

- A **trapezoid** has one pair of parallel sides, called the **bases**. The **height** meets the bases at a square corner.
- **Decompose**: cut it into a rectangle and one or two triangles. Find each area, then **add**.
- The triangle bases together equal the **long base – short base**. Two matching triangles split it in half.

WORKED EXAMPLE

Bases 6 cm and 14 cm, height 5 cm, two matching triangles.

Step 1 Rectangle: $6 \times 5 = 30 \text{ cm}^2$.**Step 2** Each triangle base: $(14 - 6) \div 2 = 4 \text{ cm}$.**Step 3** Triangles: $2 \times (\frac{1}{2} \times 4 \times 5) = 20 \text{ cm}^2$.**Answer** $30 + 20 = 50 \text{ cm}^2$ **WORDS** trapezoid (*trapezio*) · decompose (*descomponer*) · height (*altura*)

PRACTICE

show your work · label units

1. Each trapezoid is cut into a rectangle and two matching triangles. Find the base of one triangle.

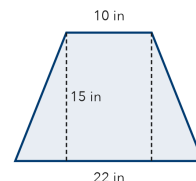
a. bases 10 in and 22 in _____

b. bases 8 m and 18 m _____

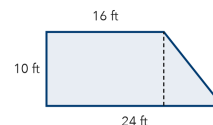
c. bases 12 cm and 20 cm _____

d. bases 17 m and 27 m _____

- 2.
- Craft Tabletop.**
- A craft tabletop is an isosceles trapezoid. It is cut into a rectangle and two matching triangles.

a. Area of the rectangle _____ in²b. Area of both triangles _____ in²c. Area of the tabletop _____ in²

- 3.
- Right-Angle Deck.**
- A deck is a trapezoid with one square corner. Decompose it into a rectangle and a triangle. Find its area.

Hint One triangle only. Its base is the long side minus the short side.

Show your work.

Answer _____ ft²

- 4.
- Bus Shelter Roof.**
- A roof panel is an isosceles trapezoid with bases 8 m and 18 m and a height of 6 m. Find its area.

Hint Split the leftover $18 - 8$ into two equal triangle bases.

Show your work.

Answer _____ m²

- 5.
- Corner Garden.**
- A garden is a trapezoid with one square corner. Its parallel sides are 11 yd and 19 yd, and the height is 13 yd. Find its area.

Show your work.

Answer _____ yd²

